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ENTERPRISE SOLUTION DOCUMENT

HRMS Operations Enterprise Platform

Enterprise Architecture & Comprehensive Workflow Specification

An end-to-end operational guide explaining the role-based workflows, automated pipelines, talent acquisition lifecycle, payroll processing, and self-service capabilities for executive clients and stakeholders.

CLIENT & AUDIENCE	TECHNOLOGY STACK	PLATFORM VERSION
Executive Stakeholders & HR Leadership	Java 23 • Spring Boot 3.3.3 • MySQL 8.0	Enterprise SaaS v2.4 (Production Ready)
DEPLOYMENT HOST	**SECURITY ARCHITECTURE**	**DOCUMENT STATUS**
Localhost Enterprise Engine (Port 8081)	Spring Security 6 • BCrypt • RBAC	Official Client Walkthrough Spec

01 System Architecture & Core Stack

HRMS Operations is architected as a modern, high-throughput enterprise platform engineered to unify workforce management, talent acquisition, attendance tracking, compensation runs, and organizational governance into a single, cohesive SaaS ecosystem.

LAYER	TECHNOLOGY CHOSEN	ARCHITECTURAL ROLE & ENTERPRISE BENEFIT
Core Engine	Java 23 (LTS) & Spring Boot 3.3.3	High-performance virtual threads, strict memory management, and production-grade REST/MVC controllers.
Database Layer	MySQL 8.0 & Hibernate 6 / JPA	ACID compliance, relational foreign-key integrity, HikariCP connection pooling, and optimized indexing.
Security & Auth	Spring Security 6 & BCrypt	Role-Based Access Control (RBAC), multi-identifier user authentication (Username, Email, or Employee Code).
Presentation	JSP + Enterprise Tailwind Tokens	Server-rendered responsive UI with an 8px grid, dark slate navigation, live notifications, and interactive popups.

02 Personas & Role-Based Access Matrix

The platform enforces strict role-based authorization to guarantee privacy, statutory compliance, and separation of duties.

A System Administrator

ROLE_ADMIN

Full operational governance, platform auditing, organization hierarchy configuration, role assignments, and system-wide overrides.

`Username: admin  Password: admin123`

H HR Manager (Elena Rostova / Sarah)

ROLE_HR_MANAGER

Talent acquisition Kanban pipeline, candidate hiring, onboarding checklists, leave approvals, employee lifecycle and directory management.

`Username: hrmanager  Password: hr123`

P Payroll Manager (Marcus Vance)

ROLE_PAYROLL_MANAGER

Salary structure definitions, batch compensation runs, attendance & overtime calculation, tax deduction formulas, and payslip distribution.

`Username: payroll  Password: payroll123`

M Engineering Lead / Manager (David Lee)

ROLE_MANAGER

Team supervision, direct report leave authorizations, attendance regularizations, candidate technical interviews, and KPI reviews.

`Username: david  Password: user123`

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General Employee (Priya Nair / Alex)

ROLE_EMPLOYEE

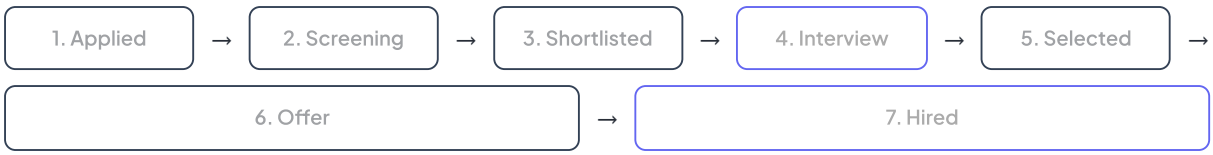
Dedicated Employee Self-Service (ESS) hub: real-time punch clock, leave quota tracking, attendance history, onboarding tasks, and payslips.

`Username: priya  Password: user123`

03 Talent Acquisition & Onboarding Lifecycle

The recruitment module integrates applicant tracking directly with employee provisioning. Candidates advance through structured stages until hiring automatically initiates the cross-department onboarding engine.

RECRUITMENT PIPELINE STAGES



Step 1: Requisition & Sourcing

HR creates job postings specifying department, vacancies, closing dates, and hiring manager. Candidates register with contact information, experience, and expectations.

Step 2: Technical & Managerial Interviews

Interviews are scheduled with calendar times and secure meeting URLs. Assigned interviewers submit quantitative scores (1 to 5) and qualitative evaluation notes directly on the platform.

Step 3: One-Click Hiring & Employee Auto-Provisioning

When the candidate is moved to `HIRED`, the system triggers a transactional creation workflow: generating an employee code (`EMP-100X`), linking reporting managers, initializing leave balances, creating salary structures, and generating initial login credentials.

Step 4: Cross-Functional Onboarding Checklist

The system auto-generates department-specific onboarding items across HR (document verification), IT (hardware provisioning & credentials), Manager (1-on-1 alignment), and Employee (compliance training).

04 Employee Self-Service (ESS) Portal

Employees access a tailored dashboard designed for daily productivity and transparency. Self-service reduces HR administrative burden while keeping workers informed.

DAILY EMPLOYEE SELF-SERVICE ROUTINE



Live Attendance Punch Clock

Employees punch in upon arrival. The system computes standard work hours (8h), flags late arrivals after 09:30 AM, logs overtime, and allows punch-out upon day completion.

Leave Quota Cards

Interactive cards display total, consumed, and remaining allowances for Annual, Casual, and Sick leaves for the current calendar year with instant application submission.

Attendance Correction Requests

If an employee misses a punch or encounters transit delays, they submit an attendance regularization note with requested times for manager review.

Confidential Payslip Access

Employees securely review and download their monthly compensation breakdowns, showing allowances, statutory deductions, overtime bonuses, and net take-home pay.

05 Attendance Engine & Leave Governance

The platform enforces rigorous operational rules to ensure fair timekeeping and automated policy compliance.

Standard Working Hours & Late Threshold

Standard workday is calculated at **8.0 hours**. Any check-in registered past **09:30 AM** automatically logs late minutes and tags the record status as `LATE` for supervisor visibility.

Overtime Calculation Formula

Time recorded in excess of the standard 8.0 hours is tracked as overtime hours. During monthly compensation processing, approved overtime is dynamically calculated and added to the employee's payroll allowances.

Leave Approval & Overlap Protection Workflow

The leave engine automatically excludes Saturdays and Sundays from leave calculations, prevents overlapping date bookings, and routes requests to the reporting manager with instant in-app alerts.

06 Automated Payroll & Compensation Cycle

The payroll module provides mathematical precision, duplicate protection, and statutory compliance.

MONTHLY PAYROLL GENERATION CYCLE



COMPONENT	TYPE	CALCULATION RULE
Basic Salary	Earnings	Contracted monthly fixed base compensation.
House Rent Allowance (HRA)	Earnings	Configurable allowance (default 40% of Basic Salary).
Special / Conveyance / Medical	Earnings	Fixed allowances for transport, healthcare, and performance.
Overtime Pay	Variable Earning	Accumulated approved overtime hours × hourly wage rate.
Provident Fund (PF)	Statutory Deduction	12% statutory deduction on Basic Salary.
Professional Tax & Income Tax	Statutory Deduction	Configurable progressive deduction brackets.
Net Take-Home Pay	Total Disbursed	`(Basic + Total Allowances) - Total Statutory Deductions`

07 In-App Real-Time Notification & Popup Toast System

To ensure critical HR events are never overlooked, the platform incorporates a floating notification toast system across the entire application shell.

Automated Notification Triggering

Notifications are automatically dispatched when leave requests are submitted, interviews are scheduled, candidates accept offers, attendance corrections are requested, or compensation runs are completed.

Interactive Floating Popup Card

When an authenticated user (Admin, HR, Manager, or Employee) opens their dashboard or portal, unread notifications appear as animated floating cards on the lower-right corner of the workspace.

Instant Dismissal & "Mark as Read"

Each popup card provides an **Open** link and a **Mark as Read** button. Clicking "Mark as Read" makes an asynchronous background call to the database, decrements the topbar bell badge, and animates the card off the screen.

08 Client Demo Verification Guide

Stakeholders can verify all workflows on the live local deployment:

MODULE NAME	DIRECT ACCESS URL	RECOMMENDED TEST ACCOUNT	KEY ACTIONS TO SHOWCASE
Executive Dashboard	`/dashboard`	`admin` (pass: `admin123`)	Live headcount, 7-day attendance trends, department distribution, quick actions.
Self-Service Portal	`/portal`	`priya` (pass: `user123`)	Live punch clock, leave quota balances, task list, personal payslips.
Recruitment ATS	`/recruitment`	`hrmanager` (pass: `hr123`)	7-stage candidate Kanban, interview scheduling, feedback ratings, hire candidate.
Leave Management	`/leave`	`david` (pass: `user123`)	Pending leave requests, approve/reject workflow, quota deductions.
Payroll & Salary	`/payroll`	`payroll` (pass: `payroll123`)	Batch payroll calculation, duplicate run protection, detailed printable payslips.
AI HR Insights	`/insights`	`admin` (pass: `admin123`)	Automated turnover risk analysis, absenteeism detection, operational alerts.